

# HAO SUN

+86-17863107076 | [sunhaoxx@zju.edu.cn](mailto:sunhaoxx@zju.edu.cn) | [kiva12138.github.io](https://kiva12138.github.io) (personal page)

 [Hao Sun](#) |  [Kiva12138](#) |  0000-0001-8094-1991 |  [Kiva12138](#)

## OBJECTIVE

**Ph.D.** in Computer Science and Technology from **Zhejiang University**, with previous experience as a **Senior Researcher** and Visiting Scholar at Ritsumeikan University, Japan. Research focuses on **Multimodal Learning, Large Language Models (LLMs), Vision-Language-Action (VLA) models, Embodied AI, Reinforcement Learning, and Affective Computing**, dedicated to advancing Artificial General Intelligence (AGI). Published multiple first-author papers in top-tier international conferences and journals, including ACL, ACM Multimedia, Information Fusion, IEEE Transactions on Affective Computing, and Pattern Recognition, accumulating over 500 citations alongside several patents. Extensively involved in National Key R&D Programs, provincial natural science funds, and industry-academia collaboration projects, demonstrating robust scientific research and engineering practice capabilities. Recipient of prestigious honors, including the Outstanding Graduate of Zhejiang University and the Huawei Scholarship, combining academic innovation with exceptional project organization and coordination skills.

## EDUCATION

- **Zhejiang University**  09.2020 - 06.2025  
Ph.D in Computer Science and Technology Hangzhou, China  
◦ Awarded the Outstanding Graduate Honor, selected as one of the top 10% of graduates for academic achievement.
- **Harbin Institute of Technology**  09.2016 - 06.2020  
Bachelor of Software Engineering Harbin, China  
◦ Awarded the Outstanding Graduate Honor, granted to the top 8% of students.

## EXPERIENCE

- **Ritsumeikan University, College of Computer Science and Engineering**  06.2025 - Now  
Senior Researcher, invited by Yen-Wei Chen, Fellow of the Engineering Academy of Japan Osaka, Japan  
◦ Oversaw the research part of **LLM, Multimodality, VLA, and Embodied AI** in the host laboratory.  
◦ Led a research team focused on **AGI and VLA** utilizing LLMs, reinforcement learning, and bionics.
- **Ritsumeikan University, College of Computer Science and Engineering**  08.2023 - 08.2024  
Visiting Scholar, invited by Yen-Wei Chen, Fellow of the Engineering Academy of Japan Osaka & Otsu, Japan  
◦ Funded by the Zhejiang University PhD Academic Star Program (awarded to the top 100 graduate students).  
◦ Led a project on developing a unified multimodal and multitask framework with LLMs.  
◦ Led a project on enabling LLMs with multimodal processing capabilities through parameter-efficient fine-tuning.  
◦ Published findings at IEEE Transactions on Affective Computing, Pattern Recognition, etc.

## PROJECTS

- **Intelligent Integrated Analysis Platform Construction for Rheumatoid Arthritis (RA)** 2022 - 2025  
Funded by National Key R&D Program Project, 2022YFC2504605, Ministry of Science and Technology, China Core Participant  
◦ Aimed to develop an AI-driven platform for integrated analysis to enhance diagnosis and treatment of RA.  
◦ Built a new approach that integrates multimodal clinical data and optimizes diagnosis accuracy by 10%.  
◦ Accountable for the development, implementation, and validation of multimodal methodologies.
- **Preoperative Early Recurrence Detection and Prediction of HCC Based on Federated Learning** 2022 - 2024  
Funded by Zhejiang Provincial Natural Science Foundation Key Project, LZ22F020012 Core Participant  
◦ Aim to develop a federated learning solution for early preoperative HCC recurrence prediction with privacy.  
◦ Achieved +13% accuracy in recurrence prediction while ensuring data privacy through federated learning.  
◦ Responsible for project proposals, multimodal methodologies, and final validation.
- **Research on Key Technologies for Smart Construction Site Management Platform Based on CV** 2022 - 2024  
Funded by Hangzhou New Zhongda Technology Co., Ltd., 2022AIZD0147-02 Core Participant  
◦ Aimed to develop a smart construction site monitor and management platform to improve safety and efficiency.  
◦ Successfully built a real-time monitoring platform that reduced violations by 15% and simplified management.  
◦ Responsible for project proposals, methodology, acceptance, and management.

- [C.1] [First Author]. **CubeMLP: An MLP-based Model for Multimodal Sentiment Analysis and Depression Estimation**. In *ACM Multimedia Proceedings*. 2022. pp.37223729. Association for Computing Machinery. Lisboa, Portugal. DOI: 10.1145/3503161.3548025. Cited by 140 (as of June 2025).
- [C.2] [First Author]. **MIRTH: Mutual-Information Reasoning with Temporal Hubs for Vision-Language-Action Agents**. Expected to publish in *The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)*.
- [J.1] [First Author]. **Modality-Invariant Temporal Representation Learning for Multimodal Sentiment Classification**. *Information Fusion*. 2023. Vol.91. pp.504-514. Elsevier. DOI: 10.1016/j.inffus.2022.10.031. Cited by 40 (as of June 2025). Impact Factor: 18.1.
- [J.2] [First Author]. **Tensorformer: A tensor-based multimodal transformer for multimodal sentiment analysis and depression detection**. *IEEE Transactions on Affective Computing*. 2023. Vol.14(4). pp.2776-2786. IEEE. DOI: 10.1109/TAFFC.2022.3233070. Cited by 45 (as of June 2025). Impact Factor: 13.9.
- [J.3] [First Author]. **Multimodal Sentiment Analysis with Mutual Information-based Disentangled Representation Learning**. *IEEE Transactions on Affective Computing*. 2025. Vol.16(3). pp.1606-1617. IEEE. DOI: 10.1109/TAFFC.2025.3529732. Impact Factor: 13.9.
- [J.4] [First Author]. **One Framework to Rule Them All: Unifying Multimodal Tasks with LLM Neural-Tuning**. *Pattern Recognition*. 2025. Early Access. Elsevier. DOI: 10.1016/j.patcog.2025.112275. Impact Factor: 8.5.
- [J.5] [First Author]. **Multi-Modal Adaptive Fusion Transformer Network for The Estimation of Depression Level**. *SENSORS*. 2021. Vol.21(14). pp.4764. MDPI. DOI: 10.3390/s21144764. Cited by 80 (as of June 2025).
- [S.1] [First Author]. **Multimodal Infusion Tuning for Large Models**. *ArXiv.2403.05060*, 2024.
- [S.2] [First Author]. **Robust Latent Representation Tuning for Image-text Classification**. *ArXiv.2406.06048*, 2024.
- [S.3] [First Author]. **Modality-invariant and Specific Prompting for Multimodal Human Perception Understanding**. *ArXiv.2311.10791*, 2023.
- [C.3] [Co-First Author]. **EPIC: Efficient Prompt Interaction for Text-Image Classification**. In *IEEE International Conference on Multimedia & Expo (ICME)*. 2025, pp. 1-6. IEEE. Nantes, France. DOI: 10.1109/ICME59968.2025.11210028.
- [C.4] [Second Author]. **IRLSG: Invariant Representation Learning for Single-Domain Generalization in Medical Image Segmentation**. In *IEEE International Conference on Acoustics, Speech and Signal Processings (ICASSP)*. 2024. pp.5585-5589. IEEE. DOI: 10.1109/ICASSP48485.2024.10446700.
- [C.5] [Second Author]. **CG-DMER: Hybrid Contrastive-Generative Framework for Disentangled Multimodal ECG Representation Learning**. In *IEEE International Conference on Acoustics, Speech and Signal Processings (ICASSP)*. 2026. IEEE.
- [C.6] [Second Author]. **Improving scDiffusion with Sparsity-Biased Classifier-Free Guidance**. In *IEEE Engineering in Medicine and Biology Society*. 2026. Accepted.
- [C.7] [Third Author]. **Dynamic Summary Generation for Interpretable Multimodal Depression Detection**. In *IEEE International Conference on Acoustics, Speech and Signal Processings (ICASSP)*. 2026. IEEE.
- [C.8] [Third Author]. **LGA: A Language Guide Adapter for Advancing the SAM Models Capabilities in Medical Image Segmentation**. In *International Conference on Medical Image Computing and Computer-Assisted Intervention Processings (MICCAI)*. 2024. pp.610-620. Springer. DOI: 10.1007/978-3-031-72390-2\_57.
- [C.9] [Third Author]. **MCKD: Mutually Collaborative Knowledge Distillation for Federated Domain Adaptation And Generalization**. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2023. pp.1-5. IEEE. DOI: 10.1109/ICASSP49357.2023.10095699.
- [J.6] [Third Author]. **CoSTHR: A Heart Rate Estimating Network with Adaptive Color Space Transformation**. *IEEE Transactions on Instrumentation and Measurement*. 2022. Vol.71. pp.1-10. IEEE. DOI: 10.1109/TIM.2022.3170976. Cited by 25 (as of June 2025).
- [J.7] [Third Author]. **DepressionLLM: Emotion- and Causality-aware Depression Detection with Foundation Models**. *Displays*. 2025. Vol.92. pp.103304. Elsevier. DOI: j.displa.2025.103304.
- [C.10] [Third Author]. **Enhanced Multimodal Depression Detection With Emotion Prompts**. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2025. pp.1-5. IEEE. DOI: 10.1109/ICASSP49660.2025.10889035.
- [P.1] **Engineering Progress Determining Method and Device Based on Multi-Mode Time Sequence Information Fusion**. App.No: US20250005475A1. Country: CN/US. Publication Date: 2023-07.
- [P.2] **Engineering Progress Determining Method and Device Based on Multi-Mode Time Sequence Information Fusion**. Grant No: CN116502882B. Country: CN. Grant Date: 2023-10.

[P.3] **Rheumatoid Arthritis Activity Grading Device Based on Multimodal Data.** Grant No: CN116797572B.  
Country: CN. Grant Date: 2025-09.

[P.4] **A Single-Domain Generalization Method for Medical Image Segmentation.** Grant No: CN116596832B.  
Country: CN. Grant Date: 2025-07.

[R.1] **MIRTH Robot Control System V1.0** Registration No.: 2026SR0601586, China, Authorization Date: May 2026.

KEYNOTES

- **Embodied Empathy: Empowering AI with Perception, Cognition, Action, and Empathy.**  
International Science and Technology Innovation Talents’ China Tour  
Invited by the International Talent Exchange Center of the Ministry of Science and Technology.  
04.2026, Suzhou and Beijing.
- **Multimodal contrastive learning enhancement methods for image classification and segmentation scenarios.**  
International Workshop on Computer Vision and Artificial Intelligence  
Invited by Yen-Wei Chen, Academician of the Engineering Academy of Japan.  
02.2024, Kusatsu, Japan.

HONORS AND AWARDS

- **Excellent Postgraduate Students’ Award** 06.2025  
Zhejiang University Hangzhou, China
  - Awarded to the top 10% of outstanding doctoral students in recognition of their academic excellence.
- **Award of Honor for Graduate (Four times)** 12.2024 / 12.2023 / 12.2022 / 12.2021  
Zhejiang University Hangzhou, China
  - Awarded annually to the top 15% of outstanding doctoral students in recognition of their excellence.
- **Zhejiang University Academic Scholarship** 12.2022  
Zhejiang University Hangzhou, China
  - Awarded to support research by outstanding doctoral students.
- **Outstanding Graduate Leader Award (Twice)** 12.2024 / 12.2023  
Zhejiang University Hangzhou, China
  - Recognizes exceptional graduate students who demonstrate outstanding leadership to their field or community.
- **Graduate with Merit A Performance** 12.2023  
Zhejiang University Hangzhou, China
  - Awarded to graduates demonstrated exceptional academic performance and active participation in social activities.
- **HUAWEI Scholarship** 12.2023  
Zhejiang University Hangzhou, China
  - Awarded to exceptional students in computer science and AI for academic excellence and research innovation.
- **Outstanding Undergraduate Award** 06.2020  
Harbin Institute of Technology Harbin, China
  - Awarded to the top 15% of outstanding under-graduate students in recognition of their excellence.
- **National Aspirational Scholarship** 12.2018  
Harbin Institute of Technology Harbin, China
  - Awarded to the top 5% of outstanding undergraduate students in recognition of their excellence.

SKILLS

- **Academic-Specific:** Scholarly Writing, Publication, Peer Review, Conference Presentation, Grant Proposal
- **AI Research:** Algorithm Development, Model Training & Finetuning, Data Processing, Evaluation
- **LLM-Specific:** Model Customization, Knowledge Integration, Multimodal Tuning, Scalability and Efficiency
- **Embodied AI:** Action Grouding, Platform Building, Hardware Employment
- **Multimodal Research:** Framework Design, Task Adaptation, Multimodal System Deployment
- **Software Engineering:** Feasibility and Requirements Analysis, System and Detailed Design, Implementation and Software Maintenance, etc
- **Programming Languages:** Python, PyTorch, Numpy, TensorFlow, Java, C++, C, HTML, GO, etc
- **Full-stack Development:** FrontEnd Programming, BackEnd Design and implementation, Database Systems
- **Language:** Mandarin, English, Japanese

## ADDITIONAL INFORMATION

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2020 - 2021: Published TensorFlow tutorials on IMOOC.

2016 - 2025: Developed 30+ independent projects (please refer to <https://github.com/kiva12138>).

2026 - Now: Served as a Guest Editor for the journal Frontiers in Neuroinformatics.

2026 - Now: Served as a Guest Editor for the journal Journal of Visualized Experiments.

2022 - Now: Served as Reviewers of Information Fusion, IEEE Transactions on Affective Computing, Pattern Recognition, ACM Transactions on Multimedia Computing, Communications and Applications, IEEE/CAA journal of Automatica Sinica, Information Processing and Management, Neurocomputing, etc.

## REFERENCES

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1. **Yen-Wei Chen**

Professor, College of Information Science and Engineering, Ritsumeikan University

**Fellow of the Engineering Academy of Japan**

Email: chen@is.ritsumei.ac.jp

*Relationship: Advisor*

2. **Lanfen Lin**

Professor, College of Computer Science and Technology, Zhejiang University

**Deputy Director of the Institute of Artificial Intelligence in Zhejiang University**

Email: llf@zju.edu.cn

*Relationship: Advisor*